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**Optimization of an interplanetary trajectory design with an
intermediate Gravity Assist**

Orbital Mechanics

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Contents

1	Interplanetary Transfer	1
1.0.1	Assumptions	1
1.0.2	Global Optimization Strategy: 3D Grid Search	1
1.0.3	Grid Refinement via Phasing Analysis	3
1.0.4	Global Grid Search for Validation	4
1.0.5	Local Refinement and Gradient-Based Optimization	4
1.0.6	Detailed Post-Processing: Trajectory Reconstruction	6

1| Interplanetary Transfer

This study analyzes a trajectory from Jupiter to a high-inclination Near-Earth Object (NEO) within the 2030–2060 window. Due to the prohibitive ΔV of a direct transfer, the mission profile implements a mandatory Mars Gravity Assist. This maneuver is optimized to adjust both heliocentric energy and orbital plane, significantly improving mission efficiency.

1.0.1 Patched Conics Method Application

The analysis employs the *Patched Conics* approximation, modeling interplanetary legs as Keplerian arcs via *Lambert's Problem*. The Mars encounter is treated as a "Point-like Flyby" in the heliocentric frame, assuming negligible time within the SOI. In the planetocentric frame, the flyby is explicitly reconstructed: asymptotic velocities ($\vec{v}_\infty^-$, $\vec{v}_\infty^+$) are matched, and any required velocity change exceeding the natural gravitational deflection is provided via an impulsive maneuver (ΔV_{GA}) at the pericenter. Terminal phases at Jupiter and the Asteroid are not modeled.

1.0.1 Assumptions

To maintain computational efficiency suitable for preliminary mission design, the following simplifying assumptions are adopted:

1. **Point-Mass Approximation:** Celestial bodies are modeled as point masses. Perturbations such as non-spherical gravity harmonics (e.g., J_2), third-body gravitational effects, and solar radiation pressure (SRP) are neglected.
2. **Impulsive Maneuvers:** Propulsive maneuvers, including the Powered Gravity Assist at the pericenter, are modeled as instantaneous changes in velocity (ΔV), neglecting finite burn durations and associated gravity losses.
3. **Instantaneous Flyby:** Consistent with the *Patched Conics* approximation, the time spent within the Sphere of Influence is considered negligible relative to the heliocentric transfer time. Consequently, the position of the spacecraft during the flyby is assumed to coincide with the planet's center of mass at the epoch t_{GA} .

1.0.2 Global Optimization Strategy: 3D Grid Search

To identify optimal launch windows and define the "basin of attraction" for local optimization, a global combinatorial search strategy (Brute-Force Grid Search) was implemented. The algorithm discretely explores the intrinsically three-dimensional solution space, defined by departure (t_{dep}), flyby (t_{ga}), and arrival (t_{arr}) dates, by constructing a cost tensor $\mathbf{J} \in \mathbb{R}^{N_{dep} \times N_{ga} \times N_{arr}}$. Each element represents the total ΔV for a specific date combination, allowing the identification of the global optimum by extracting the indices (i^* , j^* , k^*) that minimize the scalar objective function:

$$\min_{i,j,k} \mathbf{J}(i, j, k) = \min_{i,j,k} \left(\Delta V_{dep} + \Delta V_{GA}^{powered} + \Delta V_{arr} \right) \quad (1)$$

Heliocentric Legs Modeling For each pair of dates, the interplanetary trajectories (Leg 1: Earth $\rightarrow$ Mars, Leg 2: Mars $\rightarrow$ Asteroid) are computed by solving *Lambert's Problem*. This yields the hyperbolic excess velocity vectors at the intermediate body:

- $\vec{v}_\infty^-$: Incoming hyperbolic excess velocity at Mars (end of Leg 1).
- $\vec{v}_\infty^+$: Outgoing hyperbolic excess velocity from Mars (start of Leg 2).

Constraint Handling and Feasibility Domain Operational constraints are applied to define the feasible solution space. These are handled in two distinct ways within the algorithm:

- **Hard Constraints (Filtering):** Limitations on the propulsion system capabilities are enforced by discarding infeasible solutions (assigning NaN to cost tensor **J**):
 - **Launch Capacity:** $\Delta V_{dep} \leq \sqrt{C_{3,max}}$, with a maximum characteristic energy limit corresponding to an hyperbolic excess velocity of 12.0 km/s.
 - **Arrival Capacity:** $\Delta V_{arr} \leq \Delta V_{max}$, where the onboard propellant budget limits the maximum arrival maneuver to 15.0 km/s.

Note: These upper bounds are intentionally relaxed compared to standard operational missions. This choice allows the optimization algorithm to explore a broader solution space for academic analysis, filtering out only extreme outliers without being restricted by specific hardware limitations.

- **Physical Constraints (Penalty):** The geometric constraint on the flyby pericenter radius ($r_p \geq r_{safe}$), where r_{safe} is defined including a **50 km safety buffer** above the Martian atmosphere (assumed $h_{atm} = 120$ km), is **not handled via filtering** but through the Powered Gravity Assist model described in the next paragraph. Solutions requiring a perigee below this safety threshold are geometrically clamped to r_{safe} , and the violation is converted into an equivalent propellant cost (ΔV_{GA}), naturally penalizing the objective function without reducing the search space.

Gravity Assist Model and Physical Justification A critical aspect of the algorithm lies in the computation of the intermediate maneuver cost (ΔV_{GA}). The mathematical formulation implemented in this study derives from the standard *Powered Gravity Assist* (PGA) vector model, as established in classical astrodynamics literature [1].

The model operates on the premise that any vector correction required by the mission geometry exceeding the natural gravitational deflection capacity of the planet is provided via an impulsive maneuver at the point of closest approach (pericenter). This strategy is not arbitrary but is grounded in the analysis of optimal powered swing-bys, exploiting the **Oberth Effect** to maximize propulsive efficiency. The change in specific mechanical energy ($\mathcal{E}$) of a spacecraft resulting from a velocity impulse (ΔV) is proportional to the instantaneous velocity magnitude (V): $\Delta \mathcal{E} \approx \vec{V} \cdot \Delta \vec{V}$, consequently, the theoretical optimum for modifying the hyperbolic trajectory is to apply the thrust where the orbital velocity is highest. In a hyperbolic flyby, the maximum velocity occurs strictly at the pericenter (r_p).

Therefore, the algorithm enforces a saturation logic: whenever the geometry requires a passage lower than the safety threshold, the flyby altitude is fixed at the minimum safe limit ($r_p = r_{safe}$). This maximizes V_p , thereby ensuring that the calculated ΔV_{GA} represents the minimum propellant cost required to achieve the target outgoing vector.

The cost calculation proceeds by comparing the **Required Deflection** (δ_{req}), computed as the geometric angle between the asymptotic vectors $\vec{v}_{\infty}^-$ and $\vec{v}_{\infty}^+$, against the **Maximum Natural Deflection** (δ_{grav}) achievable at the safety altitude:

$$\delta_{req} = \arccos \left(\frac{\vec{v}_{\infty}^- \cdot \vec{v}_{\infty}^+}{\|\vec{v}_{\infty}^-\| \cdot \|\vec{v}_{\infty}^+\|} \right) \quad \delta_{grav} = \arcsin \left(\frac{1}{e_{in}(r_{safe})} \right) + \arcsin \left(\frac{1}{e_{out}(r_{safe})} \right) \quad (2)$$

where $e(r_{safe}) = 1 + (r_{safe} v_{\infty}^2) / \mu$ represents the eccentricity of the hyperbola tangent to the atmospheric limit.

Based on this comparison, the ΔV_{GA} cost is computed according to two distinct physical regimes:

The ΔV_{GA} cost is evaluated according to two physical regimes based on the deflection comparison. In the **Ballistic Regime** ($\delta_{req} \leq \delta_{grav}$), the planet's gravity is sufficient for steering; the pericenter radius is tuned numerically to match δ_{req} and the cost is limited strictly to the velocity magnitude change, $\Delta V_{GA} = |v_{\infty}^+ - v_{\infty}^-|$. Conversely, in the **Powered Regime** ($\delta_{req} > \delta_{grav}$), where the required curvature exceeds gravitational capabilities, the spacecraft transits at the minimum altitude (r_{safe}) to maximize the sling effect. The angular deficit $\Delta\delta = \delta_{req} - \delta_{grav}$ is compensated propulsively at the pericenter, yielding $\Delta V_{GA} = \sqrt{V_{pin}^2 + V_{pout}^2 - 2V_{pin}V_{pout}\cos(\Delta\delta)}$.

It is worth noting that this implementation effectively converts the geometric constraint on the pericenter radius into a **penalty function** (Soft Constraint), rather than a strict filter (Hard Constraint). Unlike a purely ballistic search, which would discard any geometry requiring $r_p < r_{safe}$, this approach expands the feasible solution space. It allows the optimizer to identify "compromise solutions" where a small propulsive cost at the flyby (ΔV_{GA}) is accepted in exchange for a significant reduction in the departure or arrival impulses, potentially leading to a lower global mission cost (ΔV_{tot}).

1.0.3 Grid Refinement via Phasing Analysis

The preliminary reduction of the temporal search domain relies on identifying the system's natural periodicity. Since the mission involves two distinct transfers (Leg 1: Jupiter-Mars and Leg 2: Mars-Asteroid), the recurrence of favorable launch windows is governed by the synodic period (T_{syn}) of the respective celestial pairs. Using the analytical relation $1/T_{syn} = |1/T_1 - 1/T_2|$, where T represents the heliocentric orbital period, the synodic periods were determined as $T_{syn,1} \approx 2.24$ years for the Jupiter-Mars leg and $T_{syn,2} \approx 5.29$ years for the Mars-Asteroid leg.

For the full mission to be feasible, a "double coincidence" is required, meaning the optimal alignment for both legs must occur almost simultaneously. This resonance phenomenon is expected to repeat with a periodicity approximable by the least common multiple of the two synodic periods, estimated to be in the order of 11–12 years (corresponding to $5 \times T_{syn,1} \approx 11.2$ years and $2 \times T_{syn,2} \approx 10.6$ years).

This theoretical behavior is visually confirmed by the Phase Angle plots in Figure 1, which display the temporal evolution of the phase difference for the two legs.

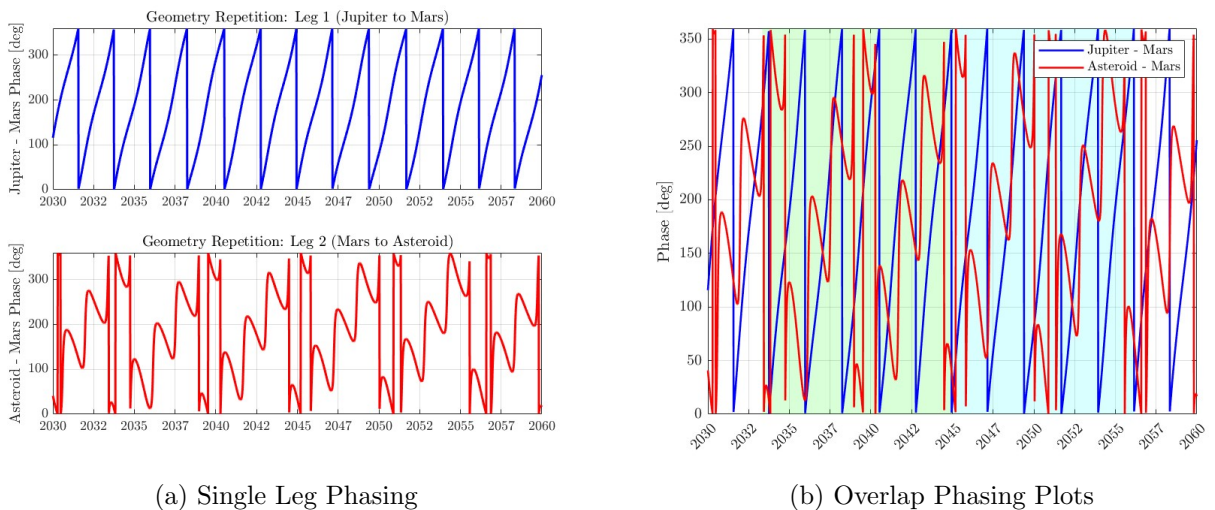


Figure 1: Phase Analysis and Pattern Synchronization

The superposition of the phasing patterns reveals two distinct macro-operational windows characterized by nearly identical geometric conditions: **Cycle 1** (2033–2044) and **Cycle 2** (2044–2056).

Notably, the ~ 11 -year interval between the onset of these two cycles provides a strong empirical validation of the theoretical resonance estimation derived from the analytical study.

1.0.4 Global Grid Search for Validation

To validate the hypothesis that these two geometrically analogous windows offered similar mission solutions, a preliminary low-resolution *Grid Search* ("Light Grid Search") was performed, focused exclusively on these two intervals. The results, visualized via the Cost Plots in Figure 2, unequivocally confirm the topological recurrence of the solution. The feasibility "islands" (low ΔV areas) appear in the same relative configuration within both cycles, validating the effectiveness of phasing analysis as a tool for predicting launch windows without the need to explore the entire temporal domain.

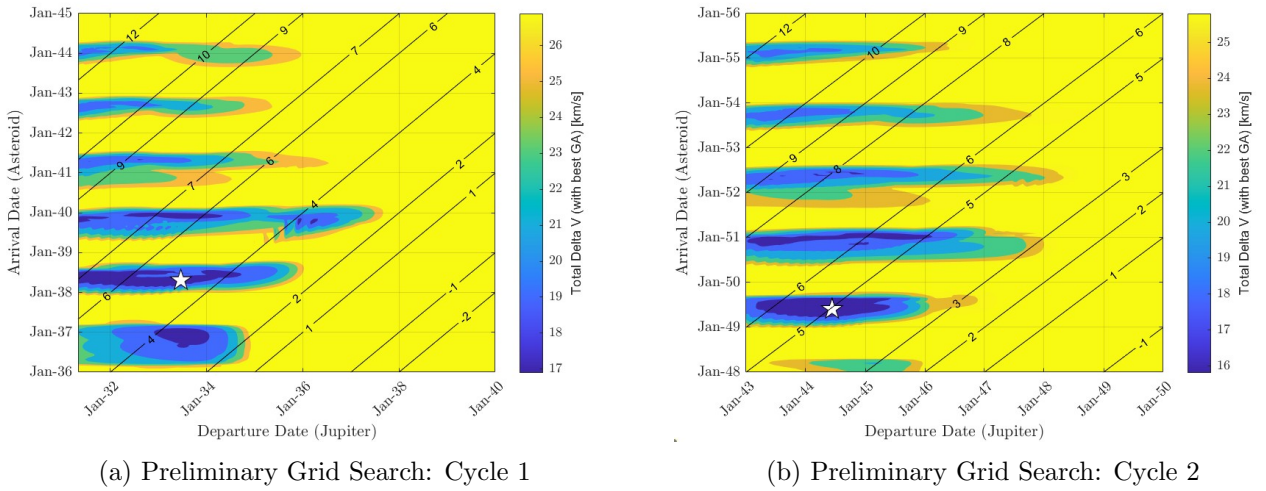


Figure 2: Comparison of low-resolution Grid Search results.

The execution of the *Global Grid Search* on the two time windows identified by the phasing analysis yielded the comparative results shown in Figure 2. Visual inspection of the Porkchop Plots unequivocally confirms the cyclic nature of the problem: the topological structure of the solutions, characterized by specific feasibility "islands" (regions of low ΔV , highlighted in dark blue), repeats with marked geometric similarity between the first cycle (2033–2044) and the second (2044–2056).

This recurrence numerically validates the synodic resonance period estimate of approximately 11 years discussed in the previous section. However, while the solution morphology is preserved, an analysis of the absolute cost values reveals significant differences due to local variations in the celestial bodies' orbital eccentricity. Comparing the global minima of the two windows, it is observed that the second time interval offers systematically more propellant-efficient solutions. Specifically, the local minima in the 2045–2056 window exhibit lower total ΔV values compared to their counterparts in the previous decade.

In light of this observation, and with the objective of minimizing the overall mission cost, the subsequent refinement and detailed optimization phases will focus exclusively on the second launch window.

1.0.5 Local Refinement and Gradient-Based Optimization

Once the most promising launch window was selected (Cycle 2: 2045–2056), a detailed analysis was conducted to pinpoint the optimal solution. The process was divided into two sequential steps:

1. **Fine Grid Search (Discrete):** A high-resolution scan (time step reduced to 3 days) was performed in the vicinity of the minimum identified by the global analysis. This step allowed for the isolation of the global minimum's "basin of attraction" and provided a robust estimate of the mission parameters (t_{dep} , t_{ga} , t_{arr}), albeit constrained by the grid's temporal discretization.

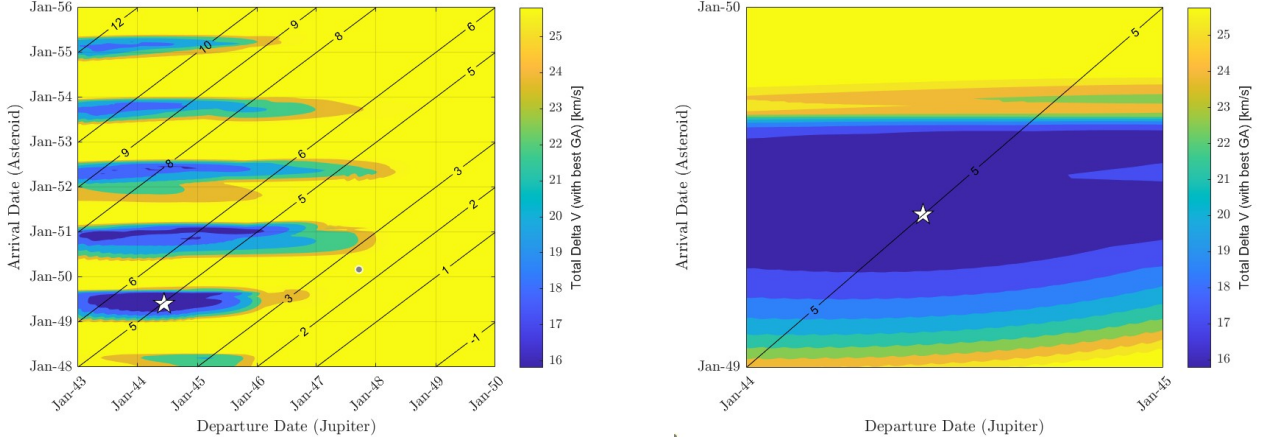


Figure 3: Fine Grid Search Cost Plot

2. **Continuous Optimization (fmincon):** The solution extracted from the Grid Search was used as the *initial guess* (x_0) for the constrained non-linear optimization algorithm **fmincon** (SQP - Sequential Quadratic Programming implementation). The optimizer, unconstrained by the discrete grid, refined the maneuver dates and flyby geometry to minimize the objective function (ΔV_{tot}) while strictly adhering to non-linear constraints (including the minimum flyby altitude at Mars).

The search space was locally restricted by imposing strict box constraints (± 60 -day buffer) around the initial guess, ensuring the solver focuses on fine-tuning the trajectory geometry without drifting toward non-physical local minima.

A distinct feature of the implementation is the hybrid management of operational limits. Insurmountable structural constraints, such as the maximum launcher capacity (C_3) and the residual ΔV at arrival, are enforced as **hard constraints** within the **missionCon** function ($c(x) \leq 0$), where strictly no violations are permitted. Conversely, the geometric limitation on the flyby altitude ($r_p \geq r_{safe}$) is handled as a **soft constraint** directly within the **missionCost** objective function.

Parameter	Fine Grid Search	Optimized (fmincon)
Departure Date	04-Jun-2044 06:32:43	13-Jun-2044 23:00:31
Flyby Date	15-Jul-2047 09:41:49	16-Jul-2047 23:32:08
Arrival Date	04-Jun-2049 20:21:49	05-Jun-2049 15:00:07
Total ΔV [km/s]	15.7766	15.7753

Table 1: Comparison between Grid Search solution and Final Optimal Solution

The results of the refinement process are summarized in Table 1. It can be observed that the optimizer produced a fine-tuning of the maneuver dates, resulting in further ΔV savings compared to the grid solution and ensuring perfect compliance with vector constraints.

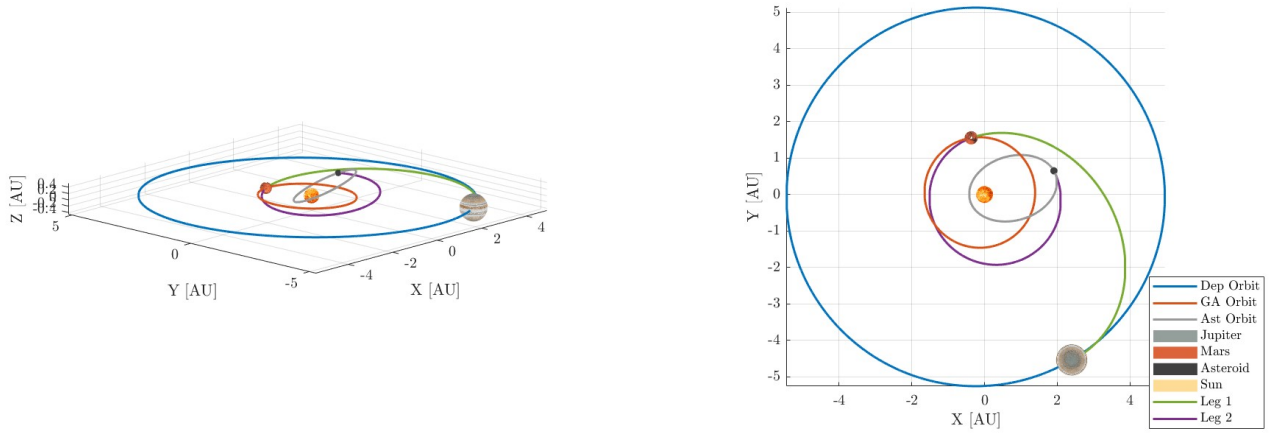


Figure 4: Complete Interplanetary Transfer

1.0.6 Detailed Post-Processing: Trajectory Reconstruction

Following the convergence of the optimization algorithm, a detailed post-processing analysis was performed to validate the physical feasibility of the flyby maneuver. The analysis transitions from the heliocentric to the planetocentric frame, using the optimal asymptotic velocities ($\vec{v}_{\infty}^{-}, \vec{v}_{\infty}^{+}$) to numerically reconstruct the exact hyperbolic trajectory.

Unlike the analytical approximation used in the global search, this phase employs a root-finding algorithm (**fzero**) to determine the precise pericenter radius (r_p) required to match the geometric deflection. In cases of **Powered Flybys** (where the ballistic solution would violate the altitude constraint), the routine reconstructs the velocity vector triangle at the safety limit (r_{safe}), calculating the precise magnitude and direction of the impulsive maneuver. This step definitively confirms the ΔV_{GA} estimate provided by the optimizer and generates the visualization of the hyperbolic passage (Figure 5).

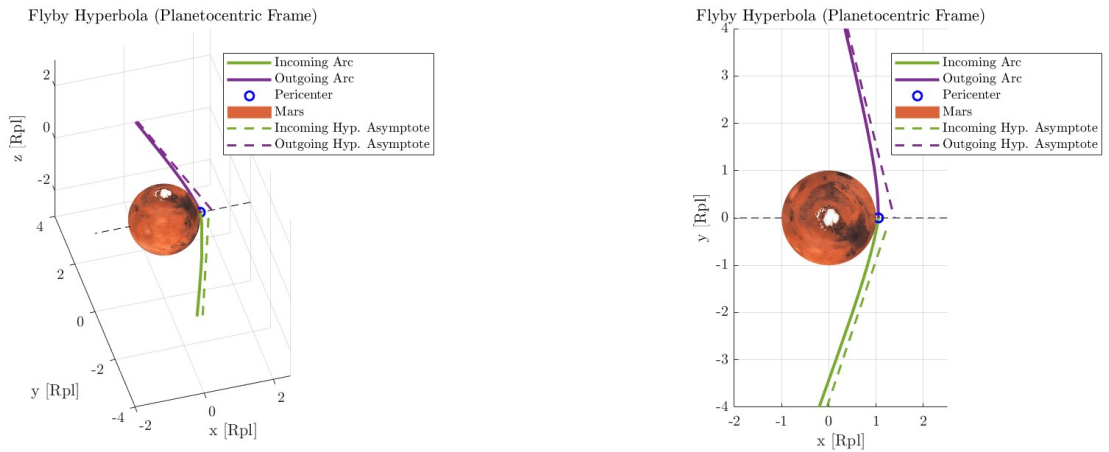


Figure 5: Gravity Assist Manoeuvre Planetocentric Plot

It is crucial to distinguish the methodology used for the trajectory reconstruction. While the heliocentric transfer arcs (Leg 1 and Leg 2) were determined by solving *Lambert's Problem* between the celestial bodies, the detailed flyby geometry within the Martian Sphere of Influence (SOI) was reconstructed analytically.

Starting from the pericenter velocity conditions (V_p) and the radius of closest approach (r_p) derived from the optimization, the incoming and outgoing hyperbolas were propagated within the planetocentric reference frame. This approach enables the visualization of the impulsive maneuver at the pericenter, ensuring vector continuity between the asymptotic velocities ($\vec{v}_\infty^-$ and $\vec{v}_\infty^+$) dictated by the heliocentric solution.

Table 2: Detailed Mars Powered Flyby Parameters

Geometric & Orbital Configuration		Dynamics, Time & Propulsion	
Pericenter Radius (r_p)	3566.0 km	Time in SOI (Δt)	45.7 h
Flyby Altitude (h_p)	170.0 km	Velocity In (V_p^{in})	9.386 km/s
Required Turn (δ_{req})	32.42°	Velocity Out (V_p^{out})	7.879 km/s
Gravity Turn (δ_{grav})	22.96°	Engine Cost (ΔV_{GA})	2.069 km/s
Engine Turn (δ_{eng})	9.46°	Total ΔV Change	4.332 km/s
Eccentricity (e_{in})	6.335	Gravitational Gain	2.262 km/s
Eccentricity (e_{out})	4.169	Impulse Ratio (η)	0.48
Semi-Major Axis (a_{in})	−668.4 km		
Semi-Major Axis (a_{out})	−1125.3 km		

Bibliography

- [1] David A. Vallado. *Fundamentals of Astrodynamics and Applications*. Microcosm Press, 2013.